PROPERTIES OF THE DIAGONAL READ OFF A PRESENTATION OF THE STACK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

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Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The description of the diagonal through a two-fibre-product square over a presentation. Used at the start of the counted proof (source lines 4094-4105). Expresses the diagonal of a presentation morphism through the diagonal of the stack.
- The criterion for checking a property of a representable morphism along a smooth surjective cover. Invoked inside the counted proof (source lines 4106-4115) as a cross-chapter pointer. Transfers properties of the diagonal between the stack and its presentation.
- The corresponding statements for properties of morphisms of algebraic spaces. Invoked at the end of the counted proof (source lines 4116-4122) as cross-chapter pointers. Supplies the equivalences at the level of the presentation.

Source reference. Original reference criteria-lemma-flat-quotient-flat-presentation: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3877–3877. Navigation pointer only: the counted text refers to the Stacks item criteria-lemma-flat-quotient-flat-presentation, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-groupoids-lemma-quotient-stack-2-cartesian: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4107–4107. Navigation pointer only: the counted text refers to the Stacks item spaces-groupoids-lemma-quotient-stack-2-cartesian, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own

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Source reference. Original reference spaces-morphisms-lemma-universally-closed-permanence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1056–1056. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-universally-closed-permanence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-check-property-covering: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 479–479. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-check-property-covering, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-section-properties-morphisms: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 248–248. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-section-properties-morphisms, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let (U, R, s, t, c) be a groupoid in algebraic spaces such that $s, t : R \to U$ are flat and locally of finite presentation. Consider the algebraic stack $\mathcal{X} = [U/R]$ (see above).*

1. *If $R \to U \times U$ is separated, then $\Delta_{\mathcal{X}}$ is separated.*
2. *If U, R are separated, then $\Delta_{\mathcal{X}}$ is separated.*
3. *If $R \to U \times U$ is locally quasi-finite, then $\mathcal{X}$ is quasi-DM.*
4. *If $s, t : R \to U$ are locally quasi-finite, then $\mathcal{X}$ is quasi-DM.*
5. *If $R \to U \times U$ is proper, then $\mathcal{X}$ is separated.*
6. *If $s, t : R \to U$ are proper and U is separated, then $\mathcal{X}$ is separated.*
7. *Add more here.*

Proof. Observe that the morphism $U \to \mathcal{X}$ is surjective, flat, and locally of finite presentation by Criteria for Representability, Lemma criteria-lemma-flat-quotient-flat-presentation. Hence the same is true for $U \times U \to \mathcal{X} \times \mathcal{X}$. We have the cartesian diagram

$$\begin{array}{ccc} R = U \times_{\mathcal{X}} U & \longrightarrow & U \times U \\ \downarrow & & \downarrow \\ \mathcal{X} & \longrightarrow & \mathcal{X} \times \mathcal{X} \end{array}$$

(see Groupoids in Spaces, Lemma spaces-groupoids-lemma-quotient-stack-2-cartesian). Thus we see that $\Delta_{\mathcal{X}}$ has one of the properties listed in Properties of Stacks, Section stacks-properties-section-properties-morphisms if and only if the morphism $R \to U \times U$ does, see Properties of Stacks, Lemma stacks-properties-lemma-check-property-covering. This explains why (1), (3), and (5) are true. The condition in (2) implies $R \to U \times U$ is separated hence (2) follows from (1). The condition in (4) implies the condition in (3) hence (4) follows from (3). The condition in (6) implies the condition in (5) by Morphisms of Spaces, Lemma spaces-morphisms-lemma-universally-closed-permanence hence (6) follows from (5). □